# MLH PE Fellowship Notes

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2 main databases + use cases:

- Relational databases: data stored in multiple, related tables. Adhere to ACID (Atomicity, Consistency, Isolation, and Durability). Usually structure data with predefined, fixed relations EXAMPLE: Oracle, MySQL, PostgreSQL

- NoSQL (Non-Relational): data does not conform to predefined schema. Store unstructured data, or semi-structured. EXAMPLE: MongoDB

4 types of NoSQL databases:

- Doc-oriented: main unit is a document which is JSON containing different fields. Good use case would be good for data analysis and looking through records

- Columnar: main unit is a column of a table, retrieved column wise. Log-base files good use case.

- Key-value: all data has a unique key identifier. Good for dealing with cache

- Graph: main unit contains nodes that represent entities and edges that represent relationships. Good from graph like structures and social networks.

Database management system (DBMS)
SQL (Structured Query Language) used on Relational DBMS

Cardinality - distinctiveness of info values contained in a column. High cardinality means column contains an outsized proportion of distinctive values. Low cardinality means more repeats. Mapping constraint is a data constraint that express the # of entities to which another entity can be related via a relationship set. For binary relationship set R on an entity A and B, there are four possible mapping cardinalities:

- One to one 1:1

- One to many 1:M

- Many to one M:1
- Many to Many M:M

Service - group of processes (known as daemons) running continuously in background.Most Linux distros uses same process manager: systemd

`Systemd` is a system + service manager for Linux, allows parallel startup of system services at boot time, on-demand activation of daemons, and dependency based on service control logic. `.service` -> system services, `.target` -> group of systemd units. Main features:

- Socket-based activation allows parallel services to start without losing messages when restarted.
- Starts services in parallel as soon as all listening sockets are in place. Significantly reduces time required to boot system
- Only stops running services on shutdown

`systemctl` - command used to control and manage `systemd` services
Activate - `systemctl start foo`, deactivate - `systemctl stop foo`, restart - `systemctl restart foo`, view status - `systemctl status foo`. More commands -> `systemctl status foo`

HTTP (hypertext transfer protocol) - set of rules for sending and receiving web pages
Web request - communicative message that goes between client/web browser to servers.
HTTP Communication happens between client and server. Client makes HTTP request and gets back a HTTP response.

HTTP request line consists of Method, URL and Version
METHODS:

- GET: used to seek info from specified source.
- HEAD: only returns headers
- POST: seeks to create and make a new object
- PUT: replace fully with new section

  URL: location that HTTP is referring to
  VERSION: what version? HTTP/3? HTTP/2? HTTP/1.1?
  HTTP request is followed by HTTP Header Structure:

- General: applied on request and response w/o affecting database body
- Request: contains info about fetched
- Response: contains location of source requested

- Entity: info about the body of resources

HTTP response has initial status line, some header lines and an entity body
Status Codes:

- 1xx: informational
- 2xx: success (200 is request was successful)
- 3xx: redirection
- 4xx: client error (404 is file not found, 400 is bad request/not in format system can read)
- 5xx: server error (500 is server error, 505 is requested HTTP version not supported)

Command: `curl http://example.org {head -silent`: curl is name of command, –head or -I tells curl to send an HTTP request with head method. And -silent tells curl to not display progress.

Caching - supplements primary database by removing unnecessary pressure on it. Different types of caching:

- Database Integrated: managed within database engine and has built in write through capabilities. The cons are size and capabilities.
- Local: stored within application, removing network traffic. Con is that each app has its own node, thus it cannot shared with other local caches.
- Remote: stored in dedicated servers by NoSQL.